

Half-Period Involutions and Exact Cancellation in Inert Lucas Sequences

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Abstract

We organize two elementary and exact cancellation mechanisms for quadratic-character sums attached to Lucas sequences modulo an inert odd prime. The first is translation by a matrix half-period, yielding the anti-periodicity relation $U_{n+T/2} = -U_n$. The second is rank-block scalar propagation, equivalently a geometric character factorization, in which the multiplier Λ^k filters complete character sums. These mechanisms arise from different structural viewpoints: the former is an odd-function pairing under index translation, whereas the latter is multiplicative filtering by the rank scalar. The determinant/norm relation links the companion-matrix determinant to the field norm of a characteristic root and forces $\text{ord}(Q) \mid T$; it is kept as an independent observation and is not itself a cancellation criterion. In the quadratic-character full-period setting, the two criteria are nevertheless related: the half-period criterion implies the rank-block scalar criterion, while the converse need not hold. No new rank-period theory is claimed.

Keywords: Lucas sequences; rank of apparition; matrix period; finite-field characters; quadratic characters; exact cancellation.

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1 Introduction

Let $(U_n)_{n \geq 0}$ be the Lucas sequence

$$U_0 = 0, \quad U_1 = 1, \quad U_{n+2} = PU_{n+1} - QU_n, \quad P, Q \in \mathbb{Z}.$$

For an odd prime p , we study complete-period sums of the form

$$\sum_n \chi_p(U_n),$$

where χ_p is the quadratic character of $\mathbb{F}_p$, extended by $\chi_p(0) = 0$.

Periodicity, ranks of apparition, and character sums for Lucas sequences have a substantial classical and modern literature; see, among others, [9, 7, 19, 4, 1, 16, 18, 12, 17, 10, 11, 15, 6, 5, 2, 13, 14, 8]. The purpose of this concise note is structural. We separate two cancellation arguments that are often used in different forms. One is a half-period involution of the companion matrix. The other is rank-block scalar propagation, or equivalently geometric character factorization, arising from the scalar multiplier of each rank block. The determinant/norm relation is kept separate as an independent observation about the matrix period. The distinction matters, especially when $\chi_p(-1) = 1$ but $\chi_p(\Lambda) = -1$. Among recent contributions, Fiebig, Mbirika, and Spilker [5] develop a period-entry-point framework for Lucas sequences modulo m , including structural relations among periods, entry points, and associated orders. The present work

considers a more specialized inert-prime setting, where the companion matrix admits a quadratic-extension interpretation. In this setting, the multiplier associated with the rank of apparition provides an additional scalar structure that leads to exact character-sum factorization. Thus, the present approach complements the period-entry-point viewpoint by exploiting multiplicative-order information available in the inert quadratic-extension framework.

The contribution of the note can be summarized as follows. We formulate two exact cancellation criteria for complete-period character sums in the inert Lucas setting — an index-translation criterion induced by the half-period involution $A^{T/2} = -I$, and a rank-block scalar criterion induced by $A^\alpha = \Lambda I$ — and we prove that, for the quadratic character over a full matrix period, the former implies the latter, while the converse fails as a cancellation argument when $\chi_p(-1) = 1$ ([proposition 5.7](#), [example 6.2](#)). The mechanisms themselves are elementary; the point is their precise separation and the resulting hierarchy.

2 Setup

Throughout, let $P, Q \in \mathbb{Z}$, let p be an odd prime, and put

$$D = P^2 - 4Q.$$

Although Lucas sequences are often studied under characteristic-zero non-degeneracy assumptions, no such global assumption is needed for the finite-field arguments below. Throughout we assume only the local conditions

$$p \nmid QD$$

and inertness, and all congruences in what follows are taken modulo p .

We use the following standard terminology.

Definition 2.1. The prime p is called *inert* for the sequence if

$$\chi_p(D) = -1;$$

equivalently, the polynomial

$$f(X) = X^2 - PX + Q \in \mathbb{F}_p[X]$$

is irreducible.

In the inert case,

$$\mathbb{F}_{p^2} = \mathbb{F}_p(\sqrt{D}),$$

and the two roots of f are

$$\lambda = \frac{P + \sqrt{D}}{2}, \quad \mu = \frac{P - \sqrt{D}}{2}.$$

They satisfy

$$\lambda + \mu = P, \quad \lambda\mu = Q, \quad \mu = \lambda^p, \quad \lambda = \mu^p.$$

The Binet formulas are

$$U_n = \frac{\lambda^n - \mu^n}{\lambda - \mu}, \quad V_n = \lambda^n + \mu^n, \tag{1}$$

where

$$V_0 = 2, \quad V_1 = P, \quad V_{n+2} = PV_{n+1} - QV_n.$$

Both sequences are regarded modulo p .

Let

$$A = \begin{pmatrix} P & -Q \\ 1 & 0 \end{pmatrix} \in \mathrm{GL}_2(\mathbb{F}_p).$$

For $n \geq 1$,

$$A^n = \begin{pmatrix} U_{n+1} & -QU_n \\ U_n & -QU_{n-1} \end{pmatrix}. \quad (2)$$

In particular, the lower-left entry of A^n is U_n . This companion-matrix viewpoint is classical; for the Fibonacci matrix modulo m , see [12].

3 Matrix Period

Define

$$T = \mathrm{ord}_{\mathrm{GL}_2(\mathbb{F}_p)}(A).$$

Since $A^T = I$, the sequences $(U_n)_{n \geq 0}$ and $(V_n)_{n \geq 0}$ are periodic modulo p , and T is a common period for both sequences. The minimal periods of the individual sequences may divide T .

Proposition 3.1 (Order of the eigenvalues). *In the inert case,*

$$T = \mathrm{ord}_{\mathbb{F}_{p^2}^\times}(\lambda) = \mathrm{ord}_{\mathbb{F}_{p^2}^\times}(\mu).$$

Consequently,

$$T \mid p^2 - 1.$$

Proof. The matrix A is diagonalizable over $\mathbb{F}_{p^2}$, with eigenvalues λ and μ . Since $\mu = \lambda^p$ and the Frobenius map $x \mapsto x^p$ is an automorphism of $\mathbb{F}_{p^2}^\times$, the two eigenvalues have the same multiplicative order. Hence the order of A is their common order. This order divides

$$|\mathbb{F}_{p^2}^\times| = p^2 - 1.$$

□

Proposition 3.2 (Half-period criterion). *For an integer m , one has*

$$A^m = -I$$

if and only if T is even and

$$m \equiv \frac{T}{2} \pmod{T}.$$

In particular, if T is even, then

$$A^{T/2} = -I \quad \text{and} \quad U_{n+T/2} = -U_n, \quad V_{n+T/2} = -V_n$$

for every nonnegative integer n .

Proof. By [proposition 3.1](#), $A^m = -I$ is equivalent to

$$\lambda^m = \mu^m = -1.$$

Since λ and μ both have order T , this occurs precisely when T is even and

$$m \equiv T/2 \pmod{T}.$$

For $m = T/2$, the identity $A^{T/2} = -I$ and the matrix formula (2) give the assertion for U_n . Alternatively, the Binet formulas (1) give both anti-periodicity relations directly. □

Remark 3.3 (Determinant and norm). Since

$$\det(A) = Q,$$

the relation $A^T = I$ implies

$$Q^T = 1 \pmod{p}, \quad \text{ord}_{\mathbb{F}_p^\times}(Q) \mid T.$$

The same relation has the field-theoretic interpretation

$$\text{Norm}_{\mathbb{F}_{p^2}/\mathbb{F}_p}(\lambda) = \lambda\lambda^p = \lambda\mu = Q.$$

Thus the determinant of the companion matrix is the norm of either characteristic root. This is an independent observation and does not itself imply a half-period involution.

4 Half-Period Translation and Determinant/Norm Observation

The half-period relation in [proposition 3.2](#) is a translation statement on indices. It should not be confused with the determinant or norm constraint in [remark 3.3](#). The latter forces the order of Q to divide the matrix period, but does not by itself imply that the period is even or that its midpoint acts as $-I$.

For a function $g : \mathbb{F}_p \rightarrow \mathbb{C}$ satisfying

$$g(-x) = -g(x),$$

the half-period relation immediately yields exact cancellation on any index set invariant under the translation $n \mapsto n + T/2$.

Proposition 4.1 (Half-period cancellation). *Assume that T is even. Let $I \subseteq \mathbb{Z}/T\mathbb{Z}$ be stable under*

$$n \mapsto n + \frac{T}{2}.$$

If $g : \mathbb{F}_p \rightarrow \mathbb{C}$ is odd, then

$$\sum_{n \in I} g(U_n) = 0 \quad \text{and} \quad \sum_{n \in I} g(V_n) = 0.$$

Proof. The translation $n \mapsto n + T/2$ partitions I into pairs. By [proposition 3.2](#),

$$g(U_{n+T/2}) = g(-U_n) = -g(U_n),$$

and the same argument applies to V_n . Summing over the pairs proves both identities. $\square$

Corollary 4.2 (Quadratic half-period cancellation). *If T is even, I is stable under $n \mapsto n + T/2$, and*

$$\chi_p(-1) = -1,$$

then

$$\sum_{n \in I} \chi_p(U_n) = 0 \quad \text{and} \quad \sum_{n \in I} \chi_p(V_n) = 0.$$

Proof. The extension $\chi_p(0) = 0$ preserves the identity

$$\chi_p(-x) = \chi_p(-1)\chi_p(x) = -\chi_p(x)$$

for every $x \in \mathbb{F}_p$. Apply [proposition 4.1](#). $\square$

5 Rank Blocks and Scalar Character Filtering

Let

$$\alpha = \min\{n \geq 1 : U_n = 0 \pmod{p}\}$$

be the rank of apparition of p , and define

$$\Lambda = U_{\alpha+1} \pmod{p}.$$

In the inert setting the set above is nonempty, so α is well defined: [lemma 5.1](#) below identifies α with the order of $\lambda/\mu \in \mathbb{F}_{p^2}^\times$. (For the existence of ranks of apparition in greater generality, see [\[3\]](#).)

Lemma 5.1 (Rank characterization). *In the inert case,*

$$U_n = 0 \pmod{p} \iff \left(\frac{\lambda}{\mu}\right)^n = 1.$$

Consequently,

$$\alpha = \text{ord}_{\mathbb{F}_{p^2}^\times} \left(\frac{\lambda}{\mu}\right).$$

Proof. Since $\lambda - \mu \neq 0$ in $\mathbb{F}_{p^2}$, the Binet formula gives

$$U_n = 0 \iff \lambda^n - \mu^n = 0 \iff \left(\frac{\lambda}{\mu}\right)^n = 1.$$

The assertion follows from the definition of α . □

Corollary 5.2. *In the inert case, $\alpha \mid p+1$.*

Proof. Since $\mu = \lambda^p$ and $\lambda\mu = Q$, one has $\lambda^{p+1} = \mu^{p+1} = Q$, hence

$$\left(\frac{\lambda}{\mu}\right)^{p+1} = 1.$$

Therefore $\text{ord}_{\mathbb{F}_{p^2}^\times}(\lambda/\mu)$ divides $p+1$, and [lemma 5.1](#) gives $\alpha = \text{ord}_{\mathbb{F}_{p^2}^\times}(\lambda/\mu)$. □

Proposition 5.3 (Rank multiplier and period decomposition). *One has*

$$A^\alpha = \Lambda I \quad \text{and} \quad \lambda^\alpha = \mu^\alpha = \Lambda.$$

If

$$\omega = \text{ord}_{\mathbb{F}_p^\times}(\Lambda),$$

then

$$T = \alpha\omega. \tag{3}$$

Moreover, for every $k \geq 0$ and $1 \leq j \leq \alpha$,

$$A^{k\alpha+j} = \Lambda^k A^j, \quad U_{k\alpha+j} = \Lambda^k U_j, \quad V_{k\alpha+j} = \Lambda^k V_j.$$

Proof. Using $U_\alpha = 0$ in (2),

$$A^\alpha = \begin{pmatrix} U_{\alpha+1} & 0 \\ 0 & -QU_{\alpha-1} \end{pmatrix}.$$

The recurrence at index $\alpha-1$ gives

$$U_{\alpha+1} = PU_\alpha - QU_{\alpha-1} = -QU_{\alpha-1},$$

so $A^\alpha = \Lambda I$. Since A^α has eigenvalues λ^α and μ^α , and equals the scalar matrix ΛI , it follows that $\lambda^\alpha = \mu^\alpha = \Lambda$.

By [lemma 5.1](#), $U_n \equiv 0 \pmod{p}$ if and only if $(\lambda/\mu)^n = 1$, and since $\alpha = \text{ord}(\lambda/\mu)$, this holds precisely when $\alpha \mid n$. Now $A^n = I$ implies $U_n = 0$ by the lower-left entry of (2), hence $\alpha \mid n$; writing $n = k\alpha$,

$$A^n = (A^\alpha)^k = \Lambda^k I,$$

which equals I exactly when $\omega \mid k$. Thus $A^n = I$ precisely when $\alpha\omega \mid n$, and (3) follows.

Finally,

$$A^{k\alpha+j} = (A^\alpha)^k A^j = \Lambda^k A^j,$$

which gives the formula for $U_{k\alpha+j}$ by comparison of lower-left entries, and the formula for $V_{k\alpha+j}$ follows from the Binet formula (1):

$$V_{k\alpha+j} = \lambda^{k\alpha+j} + \mu^{k\alpha+j} = \Lambda^k (\lambda^j + \mu^j) = \Lambda^k V_j.$$

□

Remark 5.4 (Classical context). The identity $A^\alpha = \Lambda I$ and the decomposition (3) represent the inert-case realization of classical relationships among periods, ranks of apparition, and multiplier orders for Fibonacci and Lucas sequences [12, 17, 18, 10, 11]. The divisibility $\alpha \mid p+1$ in [corollary 5.2](#) is classical as well [9, 18, 6]. Taking determinants in $A^\alpha = \Lambda I$ further gives

$$\Lambda^2 = Q^\alpha,$$

which connects the rank multiplier with the determinant/norm relation of [remark 3.3](#).

The period–entry–point framework developed by Fiebig, Mbirika, and Spilker [5] provides a complementary description of period, entry–point, and order phenomena for Lucas sequences modulo m . In the present inert prime setting, the multiplier admits an explicit finite-field interpretation through the characteristic root, and its multiplicative order gives the scalar decomposition underlying the rank-block character factorization developed below.

Proposition 5.5 (Full-period character factorization). *Let ψ be any multiplicative character of $\mathbb{F}_p^\times$, extended by $\psi(0) = 0$. For $X = U$ or $X = V$,*

$$\sum_{n=1}^T \psi(X_n) = \left(\sum_{k=0}^{\omega-1} \psi(\Lambda)^k \right) \left(\sum_{j=1}^{\alpha} \psi(X_j) \right). \quad (4)$$

In particular, if

$$\psi(\Lambda) \neq 1,$$

then

$$\sum_{n=1}^T \psi(U_n) = 0 \quad \text{and} \quad \sum_{n=1}^T \psi(V_n) = 0.$$

Proof. Partition the complete period into the blocks

$$n = k\alpha + j, \quad 0 \leq k \leq \omega - 1, \quad 1 \leq j \leq \alpha.$$

By [proposition 5.3](#),

$$X_{k\alpha+j} = \Lambda^k X_j.$$

Therefore,

$$\psi(X_{k\alpha+j}) = \psi(\Lambda)^k \psi(X_j).$$

Summing first over j and then over k proves (4). If $\psi(\Lambda) \neq 1$, the first factor is a geometric sum over a complete cycle of $\psi(\Lambda)$ and is zero. □

Thus the full-period character sum separates into an inner rank-block component and an outer scalar-orbit component. The resulting cancellation may occur entirely in the scalar-orbit direction, independently of any sign-reversing involution on the index set.

Corollary 5.6 (Quadratic scalar cancellation). *If*

$$\chi_p(\Lambda) = -1,$$

then ω is even and

$$\sum_{n=1}^T \chi_p(U_n) = 0 \quad \text{and} \quad \sum_{n=1}^T \chi_p(V_n) = 0.$$

Proof. The condition $\chi_p(\Lambda) = -1$ implies that Λ has even order (if ω were odd, then $\Lambda = (\Lambda^{(\omega+1)/2})^2$ would be a square), so ω and $T = \alpha\omega$ are even. The vanishing follows from [proposition 5.5](#), since $\chi_p(\Lambda) \neq 1$. $\square$

Proposition 5.7 (Quadratic character of the rank multiplier). *In the inert setting:*

(a) *if α is even, then $\chi_p(\Lambda) = -1$;*

(b) *if $p \equiv 3 \pmod{4}$, then $\chi_p(\Lambda) = -1$ if and only if T is even.*

In particular, whenever the half-period criterion of [corollary 4.2](#) applies to a full period, the rank-block scalar criterion of [corollary 5.6](#) applies as well; the converse need not hold ([example 6.2](#)).

Proof. (a) Put $r = \lambda/\mu$, so that $\text{ord}(r) = \alpha$ by [lemma 5.1](#), and recall that $\lambda^\alpha = \Lambda$ by [proposition 5.3](#). By Euler's criterion, since $\Lambda \in \mathbb{F}_p^\times$,

$$\chi_p(\Lambda) = \Lambda^{(p-1)/2} = \lambda^{\alpha(p-1)/2} = (\lambda^{p-1})^{\alpha/2} = (\mu/\lambda)^{\alpha/2} = r^{-\alpha/2}.$$

Since α is even and $\text{ord}(r) = \alpha$, the element $r^{\alpha/2}$ has order 2 in the cyclic group $\mathbb{F}_{p^2}^\times$, hence equals -1 ; therefore $r^{-\alpha/2} = -1$ and $\chi_p(\Lambda) = -1$.

(b) Suppose first that T is even, so that $\alpha\omega$ is even. If α is even, then $\chi_p(\Lambda) = -1$ by (a). If ω is even, then Λ cannot be a square: otherwise $\omega = \text{ord}(\Lambda)$ would divide the order $(p-1)/2$ of the subgroup of squares of $\mathbb{F}_p^\times$, which is odd when $p \equiv 3 \pmod{4}$, contradicting the evenness of ω . Hence $\chi_p(\Lambda) = -1$.

Conversely, suppose $\chi_p(\Lambda) = -1$. If ω were odd, then $\Lambda = (\Lambda^{(\omega+1)/2})^2$ would be a square, a contradiction; hence ω is even, and so $T = \alpha\omega$ is even.

The final assertion follows: on a full period, [corollary 4.2](#) requires T even and $\chi_p(-1) = -1$, that is, $p \equiv 3 \pmod{4}$, and then $\chi_p(\Lambda) = -1$ by (b), so [corollary 5.6](#) applies. $\square$

Remark 5.8 (Two mechanisms and their relation). The cancellation in [corollary 4.2](#) is a translation pairing:

$$X_{n+T/2} = -X_n.$$

It requires an appropriate index-set symmetry and an odd character, or more generally an odd test function. By contrast, the cancellation in [corollary 5.6](#) is rank-block scalar propagation, equivalently a geometric character factorization:

$$X_{k\alpha+j} = \Lambda^k X_j.$$

It requires $\psi(\Lambda) \neq 1$ and does not require $\psi(-1) = -1$. Thus scalar cancellation can occur even when the half-period negation pairing is ineffective, as in [example 6.2](#).

For the quadratic character on a full matrix period, [proposition 5.7](#) nevertheless shows that the two criteria are related: the half-period criterion implies the scalar criterion, while the converse need not hold. Moreover, the scalar criterion also forces the existence of a half-period involution: indeed, $\chi_p(\Lambda) = -1$ implies that ω is even and hence $T = \alpha\omega$ is even, so that $A^{T/2} = -I$ by [proposition 3.2](#). However, the resulting anti-periodicity produces quadratic-character cancellation only when $\chi_p(-1) = -1$.

6 Examples

Example 6.1 (Fibonacci sequence modulo 7). Take

$$(P, Q, p) = (1, -1, 7).$$

Then

$$D = P^2 - 4Q = 5,$$

which is a nonsquare modulo 7. Hence $X^2 - X - 1$ is irreducible over $\mathbb{F}_7$. The companion matrix is

$$A = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}.$$

Direct exponentiation gives

$$A^8 = -I, \quad A^{16} = I,$$

so

$$T = 16, \quad U_{n+8} = -U_n \pmod{7}.$$

The rank data provide an independent cross-check:

$$\alpha = 8 = p + 1, \quad \Lambda = U_9 \equiv -1 \pmod{7}, \quad \omega = \text{ord}_{\mathbb{F}_7^\times}(-1) = 2, \quad T = \alpha\omega = 16.$$

Here $\chi_p(-1) = \chi_p(\Lambda) = -1$, so both mechanisms are active: the half-period translation gives

$$\chi_p(U_{n+8}) = -\chi_p(U_n),$$

as in [corollary 4.2](#), and the same vanishing also follows from [corollary 5.6](#).

Example 6.2 (Inert scalar cancellation). Take

$$(P, Q, p) = (1, 2, 5).$$

Here

$$D = 1 - 8 = -7 \equiv 3 \pmod{5},$$

and 3 is a nonsquare modulo 5. Hence $X^2 - X + 2$ is irreducible over $\mathbb{F}_5$. The initial values are

$$U_0, U_1, U_2, U_3, U_4, U_5, U_6, U_7 = 0, 1, 1, 4, 2, 4, 0, 2 \pmod{5}.$$

The rank and scalar multiplier are

$$\alpha = 6, \quad \Lambda = U_{\alpha+1} = U_7 = 2.$$

Moreover,

$$\chi_p(\Lambda) = \chi_5(2) = -1, \quad \omega = \text{ord}_{\mathbb{F}_5^\times}(2) = 4, \quad T = \alpha\omega = 24.$$

Since $\alpha = 6$ is even, the relation $\chi_p(\Lambda) = -1$ is consistent with [proposition 5.7\(a\)](#). The identity $A^\alpha = \Lambda I$ reads $A^6 = 2I$, whence $A^{12} = -I$: anti-periodicity is present. However,

$$\chi_p(-1) = \chi_5(-1) = 1,$$

so the half-period translation does not itself produce quadratic-character cancellation. Instead, [proposition 5.3](#) gives

$$U_{6k+j} = 2^k U_j \pmod{5}, \quad V_{6k+j} = 2^k V_j \pmod{5}.$$

The four rank blocks have quadratic-character sums

$$3, -3, 3, -3,$$

and the alternating geometric factor $\chi_p(\Lambda)^k = (-1)^k$ makes the total vanish. This is scalar character filtering, not a negation pairing.

Remark 6.3 (Even period alone is not enough). The conditions T even and $A^{T/2} = -I$ alone do not force quadratic-character cancellation. For $(P, Q, p) = (1, 1, 5)$ one has $D \equiv 2 \pmod{5}$ (a nonsquare), rank $\alpha = 3$, multiplier $\Lambda \equiv -1 \equiv 4 \pmod{5}$, order $\omega = 2$, and period $T = 6$; then $A^3 = -I$, but $\chi_5(-1) = \chi_5(\Lambda) = 1$ and

$$\sum_{n=1}^6 \chi_5(U_n) = 4 \neq 0.$$

Together, [example 6.1](#), [example 6.2](#), and this remark realize the three configurations of the two quadratic criteria on a full period that can occur: both active, scalar only, and neither. By [proposition 5.7](#), the fourth configuration—half-period cancellation without scalar cancellation—cannot occur.

	p	$\chi_p(-1)$	$\chi_p(\Lambda)$	HP quadratic	Scalar
example 6.1	7	-1	-1	yes	yes
example 6.2	5	1	-1	no	yes
remark 6.3	5	1	1	no	no

Table 1: The three configurations of the two quadratic criteria realized by the examples. The fourth formal configuration (half-period active, scalar silent) is excluded by [proposition 5.7](#).

7 Scope Remark

The purpose of this note is to isolate cancellation mechanisms rather than to modify the classical rank–period theory. The rank–period structure supplies the algebraic scaffolding, while the contribution here is the explicit separation of translation cancellation from scalar character filtering. The decomposition into rank blocks follows from the companion-matrix identity, the definition of the rank of apparition, and the divisibility $\alpha \mid n$ whenever $U_n = 0 \pmod{p}$, which in the inert setting is an immediate consequence of [lemma 5.1](#). The structural point is the separation of two exact cancellation criteria as arguments: half-period translation and rank-block scalar propagation, equivalently geometric character factorization. For the quadratic character on a full period these criteria are related as in [proposition 5.7](#), but they remain distinct as proofs.

The hypotheses are essential. If $p \mid Q$, the companion matrix need not be invertible. If $p \mid D$, the characteristic polynomial is not separable in the required form. In either situation, the inert quadratic-extension argument used here does not apply without modification. Outside the inert setting considered here, an even matrix period alone does not in general imply that the half-period matrix is $-I$; within the inert setting, T is even if and only if $A^{T/2} = -I$, by [proposition 3.2](#).

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Data Availability

No external datasets were used. The numerical checks and examples are reproducible directly from the displayed recurrences, companion matrices, and finite-field computations; the associated verification code is archived at <https://doi.org/10.5281/zenodo.22118430>. The archived scripts `verify-examples.py` and `verify-theorem-pipeline.py` verify, for the parameter triples displayed in Section 6, the rank of apparition, the multiplier, the matrix period, the half-period identity, and the vanishing of the stated character sums.

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